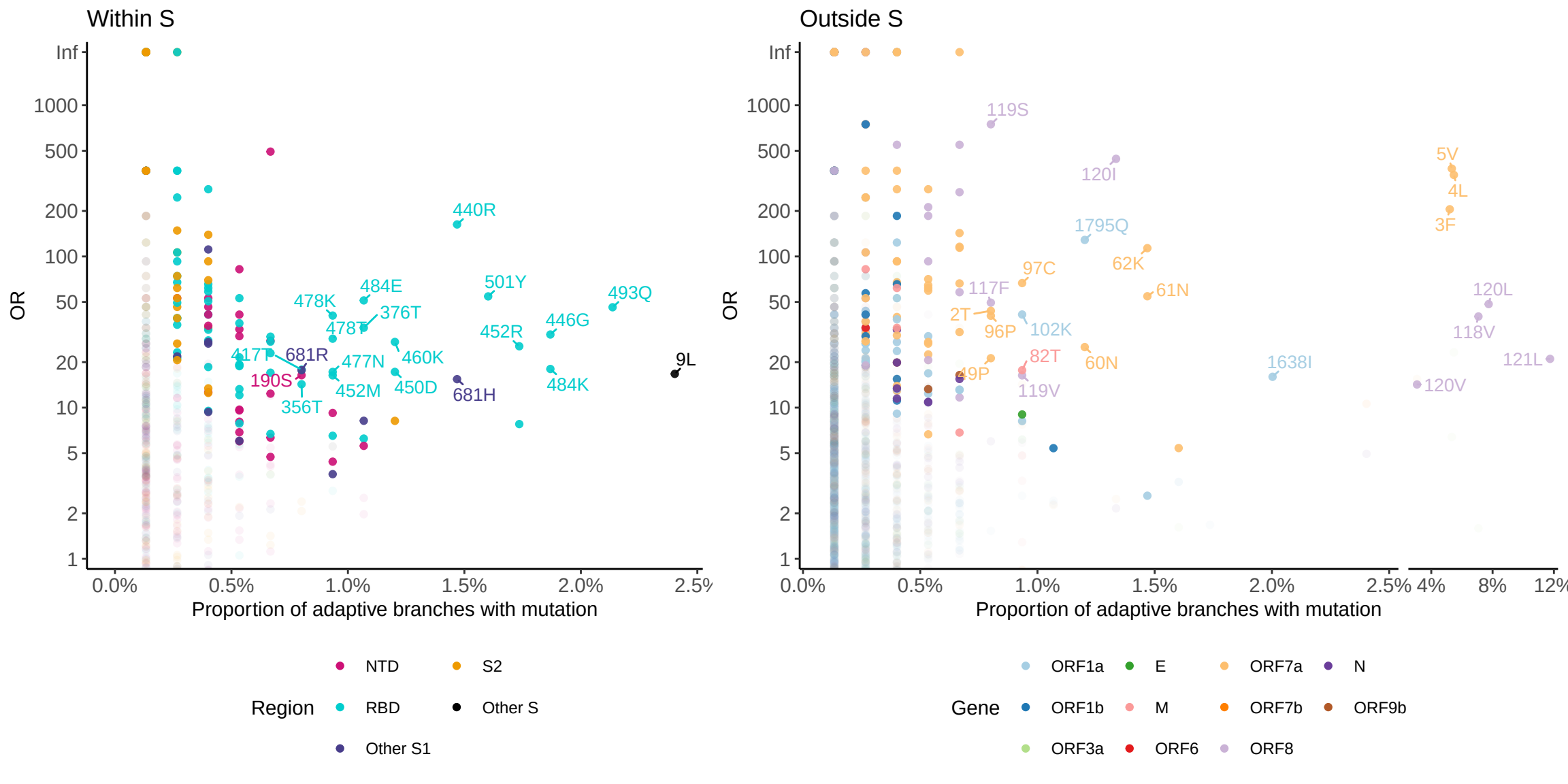

Within S
Outside S
OR
Inf
1000
500
200
100
50
20
10
5
2
1
0.0%
0.5%
1.0%
1.5%
2.0%
2.5%
4%
8%
12%
Proportion of adaptive branches with mutation
440R
501Y
493Q
484E
478K
376T
478T
446G
452R
417T
681R
477N
460K
484K
190S
452M
450D
681H
356T
9L
119S
120I
1795Q
5V
4L
3F
62K
97C
61N
117F
120L
2T
96P
102K
118V
82T
60N
1638I
121L
49P
119V
120V
Region
NTD
S2
RBD
Other S
Other S1
Gene
ORF1a
E
ORF7a
N
ORF1b
M
ORF7b
ORF9b
ORF3a
ORF6
ORF8